

G4_Task

Hello,

Please read the following carefully:

As a Software Engineering research team at the Polytechnique Montréal (Canada), we are interested in understanding developers' assessment of coding task difficulty. You will be provided with a series of 12 coding questions. Questions are divided into two parts:

- First, you will be given three instructions representing coding tasks and you will be asked to write, in a few words, what is the underlying task represented by all these instructions. The instructions represent the same tasks, but they vary in wording and information they give. In a nutshell, you have to distill the essence of the task and write it in a few words.

- In a second time, you will be asked to assess how difficult implementing this **task** would be. Note we ask the assessment based on the task you will have worded in the previous part, not based on any of the instructions presented earlier. First, you will be asked your subjective judgment on a scale from Very Easy to Very Difficult. Then, to make judging the difficulty more relatable, we adopt a similar approach as what is used in agile development and planning poker, i.e. you will be asked to evaluate how long it would take to implement a task (in minutes here). Consider in your assessment the knowledge needed to implement the task, the external information you would need to use, whether you would need the assistance of a LLM such as ChatGPT (and so the time it would take to debug the obtained code) etc. Finally, you will have to explain in a few words why you gave such difficulty ratings.

To give you an idea of the range of tasks you will need to assess, the first section of this questionnaire will be two examples we devised to illustrate the questionnaire.

Completing the task will take about 30 minutes.

Note also that all examples are written using Python 3.10 syntax.

Please make sure to answer all the questions to the best of your ability.

Please note that your identity and personal data will not be disclosed, while we plan to use the aggregated results and anonymized responses as part of a scientific publication. If you have any questions about the questionnaire or our research, please do not hesitate to contact us.

By proceeding, you consent to take part in the study under the conditions described previously.

Thank you,

Florian Tambon (florian-2.tambon@polymtl.ca)
Cyrine Zid (cyrine.zid@polymtl.ca)
Amin Nikanjam (amin.nikanjam@polymtl.ca)
Foutse Khomh (foutse.khomh@polymtl.ca)
Giuliano Antoniol (giuliano.antoniol@polymtl.ca)

* Indique une question obligatoire

1. Please enter your prolific ID : *

Example of tasks

As we ask of you some estimation of difficulty and the tasks dealt in this questionnaire can be a bit different from what you can be used to as a developer, we give you examples of some tasks along with some assessments.

Note that this assessment is **our assessment**. You may have a different opinion based on your background. The idea of those examples is to give an extreme range of what type of tasks you can expect in the questionnaire.

As a reminder: You will first be given three instructions and asked to explain what the task is about in a few words. Then, you will have to rate the difficulty of the **task** you just described (not accounting for particular instruction). Rating the difficulty is done in two steps: subjective difficulty and approximate time needed. Finally, you will be asked to say why you gave such rating.

Example 1

Instruction 1

```
1 def subtract(c1, c2):  
2     """  
3     Deliver the result of subtracting two complex numbers,  
4     specified as 'c1' and 'c2', as a complex output.  
5     :param c1: The first complex number, complex.  
6     :param c2: The second complex number, complex.  
7     :return: The difference of the two complex numbers, complex.  
8     """
```

Instruction 2

```
1 def subtract(c1, c2):
2     """
3     The method receives two complex numbers 'c1' and 'c2', from which it
4     subtracts 'c2.real' from 'c1.real' to determine the real portion, and
5     subtracts 'c2.imag' from 'c1.imag' to determine the imaginary portion.
6     It constructs and returns a new complex number using these results
7     with the 'complex()' function.
8     :param c1: The first complex number, complex.
9     :param c2: The second complex number, complex.
10    :return: The difference of the two complex numbers, complex.
11    """
```

Instruction 3

```
1 def subtract(c1, c2):
2     """
3     Subtracts two complex numbers.
4     :param c1: The first complex number, complex.
5     :param c2: The second complex number, complex.
6     :return: The difference of the two complex numbers, complex.
7     >>> complexCalculator = ComplexCalculator()
8     >>> complexCalculator.subtract(1+2j, 3+4j)
9     (-2-2j)
10    """
```

Answer

The task encapsulated by the **subtract**

function

is to compute the subtraction between two complex numbers.

This task has a subjective difficulty of **Very Easy** as assessed by us and would take about **1 min**

to implement. There is no particular difficulty, the task is very straightforward without any corner case, it just requires to know what a complex is, and the standard Python library for it.

Example 2

Context

```
1 import numpy as np
2 class MetricsCalculator2:
3     """
4     The class provides to calculate Mean Reciprocal Rank (MRR) and Mean Average Precision (MAP)
5     based on input data, where MRR measures the ranking quality and MAP measures the average precision.
6     """
7
8     def __init__(self):
9         pass
10
11     @staticmethod
12     def map(data):
13         pass
```

Instruction 1

```
1 def mrr(data):
2     """
3     Compute the MRR of the input data. MRR is a widely used evaluation index.
4     It is the mean of reciprocal rank.
5     :param data: the data must be a tuple, list 0,1, eg. ([1,0,...],5).
6     In each tuple (actual result,ground truth num),ground truth num is the total ground num.
7     ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5),([1,0,...],6),([0,0,...],5)].
8     1 stands for a correct answer, 0 stands for a wrong answer.
9     :return: if input data is list, return the recall of this list. if the input data is list of list, return the
10     average recall on all list. The second return value is a list of precision for each input.
11     >>> MetricsCalculator2.mrr([([1, 0, 1, 0], 4)])
12     >>> MetricsCalculator2.mrr([([1, 0, 1, 0], 4), ([0, 1, 0, 1], 4)])
13     1.0, [1.0]
14     0.75, [1.0, 0.5]
15
16     """
```

Instruction 2

```
1 def mrr(data):
2     """
3     Assess the Mean Reciprocal Rank (MRR) from the input 'data'.
4     MRR assesses the average of reciprocal ranks of outcomes.
5     The input structure 'data' must be a tuple consisting of a list of
6     binary values and its total count or a list of such tuples.
7     Each binary value (0 or 1) represents the correctness of an answer.
8     If 'data' is a tuple, the function returns its mean reciprocal rank.
9     If 'data' is a list, the function returns the overall average MRR along
10     with a list showing reciprocal ranks for each component tuple.
11
12     :param data: the data must be a tuple, list 0,1, eg. ([1,0,...],5).
13     In each tuple (actual result,ground truth num),ground truth num is the total ground num.
14     ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5),([1,0,...],6),([0,0,...],5)].
15     1 stands for a correct answer, 0 stands for a wrong answer.
16     :return: if input data is list, return the recall of this list. if the input data is list of list, return the
17     average recall on all list. The second return value is a list of precision for each input.
18
19     """
```

Instruction 3

```
1 def mrr(data):
2     """
3     Evaluate the Mean Reciprocal Rank (MRR) from 'data'. This metric calculates the mean of reciprocal
4     ranks for correct answers. Input 'data' could be either a tuple containing binary correctness indicators
5     and the aggregate count of answers, or multiple tuples of this type in a list. For each tuple, the function
6     measures reciprocal ranks by multiplying the binary indicators by reciprocals of their positions, finding the
7     earliest nonzero value to establish the rank of the first correct response. If the input 'data' is a tuple,
8     the MRR calculated for this specific tuple is returned; if it's a list, the function summarizes by providing
9     both the average MRR and a list of individual MRR values for each tuple.
10
11     :param data: the data must be a tuple, list 0,1, eg. ([1,0,...],5).
12     In each tuple (actual result, ground truth num), ground truth num is the total ground num.
13     ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5), ([1,0,...],6), ([0,0,...],5)].
14     1 stands for a correct answer, 0 stands for a wrong answer.
15     :return: if input data is list, return the recall of this list. if the input data is list of list, return the
16     average recall on all list. The second return value is a list of precision for each input.
17
18     """
```

Answer

The task encapsulated by the **mrr** function is to compute the Mean Reciprocal Rank of binary tuples data.

This task has a subjective difficulty of **Very Hard** as assessed by us and would take about **40 min**

to implement. The difficulty mainly branches from dealing with the tuple data and how the formula is processed accordingly (with a few corner cases). This can take additional time if the Mean Reciprocal Rank is not known.

Question 1:

Below are instructions for implementing the function.

Instruction 1

```
1 from typing import List
2
3 def concatenate(strings: List[str]) -> str:
4     """
5     Develop a function called 'concatenate' that receives 'strings',
6     a list of strings, as an argument. The function's objective is
7     to merge every string in the 'strings' list into one continuous
8     string using the .join() method. It joins all the strings in order
9     without using any delimiters and returns the concatenated string as
10    the function's outcome.
11    """
```

Instruction 2

```
1 from typing import List
2
3 def concatenate(strings: List[str]) -> str:
4     """
5     Concatenate list of strings into a single string
6     >>> concatenate([])
7     ''
8     >>> concatenate(['a', 'b', 'c'])
9     'abc'
10    """
```

Instruction 3

```
1 from typing import List
2
3 def concatenate(strings: List[str]) -> str:
4     """
5     Design a function called 'concatenate' that takes a
6     list of strings and amalgamates them into a single string.
7     """
```

2. In a few words, what is the task this function aims to implement? *

3. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

4. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

5. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 2:

Below are instructions for implementing the function:

Context

```
1 class AssessmentSystem:
2     """
3     This is a class as an student assessment system, which supports add student,
4     add course score, calculate GPA, and other functions for students and courses.
5     """
6
7     def __init__(self):
8         """
9         Initialize the students dict in assessment system.
10        """
11        self.students = {}
12
13    def add_student(self, name, grade, major):
14        pass
15
16    def get_gpa(self, name):
17        pass
18
19    def get_all_students_with_fail_course(self):
20        pass
21
22    def get_course_average(self, course):
23        pass
24
25    def get_top_student(self):
26        pass
27
28
29
```

Instruction 1

```
1 def add_course_score(self, name, course, score):
2     """
3     Insert or update the course and score for a certain student
4     within the 'self.students' dictionary. The function should
5     first confirm the presence of the student's name in the dictionary,
6     then proceed to update the course information with the new score.
7     No changes are made if the student's name does not exist in the dictionary.
8     :param name: str, student name
9     :param course: str, course name
10    :param score: int, course score
11    """
```

Instruction 2

```
1 def add_course_score(self, name, course, score):
2     """
3     Add score of specific course for student in self.students
4     :param name: str, student name
5     :param course: str, course name
6     :param score: int, course score
7     >>> system.add_student('student 1', 3, 'SE')
8     >>> system.add_course_score('student 1', 'math', 94)
9     >>> system.students
10    {'student 1': {'name': 'student 1', 'grade': 3, 'major': 'SE', 'courses': {'math': 94}}}
11    """
```

Instruction 3

```
1 def add_course_score(self, name, course, score):  
2     """  
3     Place a course score for a particular student within  
4     the 'self.students' dictionary. If that student's name is  
5     found, then proceed to update the course score accordingly.  
6     :param name: str, student name  
7     :param course: str, course name  
8     :param score: int, course score  
9     """
```

6. In a few words, what is the task this function aims to implement? *

7. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

8. How would you rate the difficulty of this task, in minutes it would take to implement?

*

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

9. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 3:

Below are instructions for implementing the function:

Instruction 1

```
1 from typing import List
2
3 def sort_numbers(numbers: str) -> str:
4     """
5     Implement a function 'sort_numbers' that receives a string 'numbers',
6     made up of words representing the numbers 'zero' to 'nine' separated
7     by spaces. The function aims to return a string where these words are
8     sorted in increasing numerical order, defined by the mapping in the
9     'to_int' dictionary that assigns each word a number from 0 to 9.
10    Initially, the function checks if 'numbers' is an empty string and
11    returns an empty string if so. If non-empty, the function splits the
12    string into individual words, sorts them by their numeric values,
13    and rejoins them into a single string, which is then returned.
14    """
```

Instruction 2

```
1 from typing import List
2
3 def sort_numbers(numbers: str) -> str:
4     """
5     Input is a space-delimited string of numerals from 'zero' to 'nine'.
6     Valid choices are 'zero', 'one', 'two', 'three', 'four', 'five',
7     'six', 'seven', 'eight' and 'nine'.
8     Return the string with numbers sorted from smallest to largest
9     >>> sort_numbers('three one five')
10     'one three five'
11     """
```

Instruction 3

```
1 from typing import List
2
3 def sort_numbers(numbers: str) -> str:
4     """
5     Create a function titled 'sort_numbers' that accepts a string
6     containing number words ('zero' through 'nine') separated by spaces.
7     The function sorts these words by their numerical equivalent in
8     ascending order. An empty string input results in an empty string output.
9     The function achieves this by setting up a dictionary that maps number
10    words to numerical values, then sorting the string if it is not empty,
11    and returning the sorted number words as a single combined string.
12    """
```

10. In a few words, what is the task this function aims to implement? *

11. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

12. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

13. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 4:

Below are instructions for implementing the function:

Context

```
1 import sqlite3
2 class StudentDatabaseProcessor:
3     """
4     This is a class with database operation, including inserting student information,
5     searching for student information by name, and deleting student information by name.
6     """
7
8     def __init__(self, database_name):
9         """
10         Initializes the StudentDatabaseProcessor object with the specified database name.
11         :param database_name: str, the name of the SQLite database.
12         """
13         self.database_name = database_name
14
15     def insert_student(self, student_data):
16         pass
17
18     def search_student_by_name(self, name):
19         pass
20
21     def delete_student_by_name(self, name):
22         pass
23
24
25
```

Instruction 1

```
1 def create_student_table(self):
2     """
3     Creates a ""students"" table in the database if it does not exist already.
4     Fields include ID of type int, name of type str, age of type int, gender
5     of type str, and grade of type int
6     :return: None
7     >>> processor = StudentDatabaseProcessor("""students.db""")
8     >>> processor.create_student_table()
9     """
```

Instruction 2

```
1 def create_student_table(self):
2     """
3     Develop a procedure that confirms or establishes a
4     'students' table in a database, equipped with ID (int),
5     name (str), age (int), gender (str), and grade (int) as fields.
6     The process should always yield a return value of 'None'.
7     :return: None
8     """
```

Instruction 3

```
1 def create_student_table(self):  
2     """  
3     Construct a functionality that ascertains the presence or creates  
4     a new 'students' table in the database, populated with the columns  
5     ID (integer), name (text), age (integer), gender (text), and grade  
6     (integer). The function is expected to return 'None' consistently.  
7     :return: None  
8     """
```

14. In a few words, what is the task this function aims to implement? *

15. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

16. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

17. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 5:

Below are instructions for implementing the function:

Instruction 1

```
1 def is_nested(string):  
2     """  
3     Define a function 'is_nested' which receives a single string  
4     composed solely of square brackets. The function aims to identify  
5     if there's any section of brackets that features nested brackets  
6     once or more, and should return True in such cases, otherwise False.  
7     """
```

Instruction 2

```
1 def is_nested(string):
2     """
3     Create a function that takes a string as input which contains only square brackets.
4     The function should return True if and only if there is a valid subsequence of brackets
5     where at least one bracket in the subsequence is nested.
6
7     is_nested('[]') → True
8     is_nested('[] [] [] [] []') → False
9     is_nested('[] []') → False
10    is_nested('[]') → False
11    is_nested('[] [] []') → True
12    is_nested('[] [] []') → True
13    """
```

Instruction 3

```
1 def is_nested(string):
2     """
3     Construct a function called 'is_nested' with a parameter 'string'
4     containing only square brackets to evaluate whether it includes a
5     segment of brackets nested to a depth of two or more. The function
6     examines each element in 'string' using a looping mechanism with index 'i'.
7     Should the element at 'string[i]' be '}', the loop skips this iteration.
8     Otherwise, it sets 'cnt' and 'max_nest' to zero, these variables tally
9     the ongoing and peak nesting depths. The next loop initiates at 'i'
10    and processes through 'string', augmenting 'cnt' for every '[' and decreasing for ']'.
11    The peak depth 'max_nest' is updated throughout this loop. When 'cnt' zeroes out
12    (signifying the end of a nested segment), if 'max_nest' is at least two,
13    True is returned to signal a suitably nested sequence. The function returns
14    False if no such sequence is discovered by the close of the loop.
15    """
```

18. In a few words, what is the task this function aims to implement? *

19. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

20. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

21. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 6:

Below are instructions for implementing the function:

Context

```
1 import json
2 import os
3 class JSONProcessor:
4     """
5     This is a class to process JSON file, including reading and writing JSON files,
6     as well as processing JSON data by removing a specified key from the JSON object.
7     """
8
9     def read_json(self, file_path):
10         pass
11
12     def write_json(self, data, file_path):
13         pass
14
15
16
```

Instruction 1

```
1 def process_json(self, file_path, remove_key):
2     """
3     Load data from a JSON file, remove a designated key,
4     and then re-save the modified data into the same file.
5     The method returns '1' if the removal and save are successful,
6     and '0' if the file cannot be read or the specific key isn't found.
7     The process starts by reading the file with 'read_json' taking 'file_path',
8     storing the outcome in 'phones'. If 'data' reports a '0' or '-1',
9     representing an unsuccessful read, then '0' is returned. Next,
10    if 'remove_key' is present in 'data', that key is deleted,
11    and the altered data is saved again using 'write_json' with 'file_path'.
12    If all goes as planned, '1' is returned. Otherwise, '0'
13    if 'removespaceremove_key' is not in 'data'.
14    :param file_path: str, the path of the JSON file.
15    :param remove_key: str, the key to be removed.
16    :return: 1, if the specified key is successfully removed and the data is written back.
17    0, if the file does not exist or the specified key does not exist in the data.
18    """
```

Instruction 2

```
1 def process_json(self, file_path, remove_key):
2     """
3     read a JSON file and process the data by removing a specified key
4     and rewrite the modified data back to the file.
5
6     :param file_path: str, the path of the JSON file.
7     :param remove_key: str, the key to be removed.
8     :return: 1, if the specified key is successfully removed and the data is written back.
9     0, if the file does not exist or the specified key does not exist in the data.
10    >>> json.read_json('test.json')
11    {'key1': 'value1', 'key2': 'value2'}
12    >>> json.process_json('test.json', 'key1')
13    1
14    >>> json.read_json('test.json')
15    {'key2': 'value2'}
16    """
```

Instruction 3

```
1 def process_json(self, file_path, remove_key):
2     """
3     Access a JSON file, remove a specified key from its content,
4     then overwrite the file with the newly modified data.
5     Return '1' if the key has been removed and the file updated successfully,
6     or '0' if the file is missing or lacks the specified key.
7     Initially, 'read_json' is utilized with 'file_path' to load the data into 'data'.
8     Should 'data' result in '0' or '-1' which denotes a reading error,
9     a return of '0' will follow. If 'data' contains 'remove_key', the key is deleted.
10    Subsequently, the data, now missing the key, is saved back to the file using 'write_json',
11    with the original 'file_path'. The method returns '1' upon a successful operation,
12    and '0' otherwise.
13    :param file_path: str, the path of the JSON file.
14    :param remove_key: str, the key to be removed.
15    :return: 1, if the specified key is successfully removed and the data is written back.
16    0, if the file does not exist or the specified key does not exist in the data.
17    """
```

22. In a few words, what is the task this function aims to implement? *

23. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

24. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

25. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 7:

Below are instructions for implementing the function:

Instruction 1

```
1 def is_sorted(lst):
2     """
3     Develop a function named 'is_sorted' that examines if a
4     list containing non-negative integers is maintained in
5     ascending sequence while ensuring no element exceeds
6     two occurrences. This function employs a dictionary
7     to monitor each integer's repetitiveness.
8     If a count surpasses two, the function halts and returns False.
9     At the end of the loop, if the list matches its sorted form,
10    the function concludes with True, provided no integer
11    has exceeded the repeat limit.
12    """
```

Instruction 2

```
1 def is_sorted(lst):
2     """
3     Given a list of numbers, return whether or not they are sorted
4     in ascending order. If list has more than 1 duplicate of the same
5     number, return False. Assume no negative numbers and only integers.
6
7     Examples
8     is_sorted([5]) → True
9     is_sorted([1, 2, 3, 4, 5]) → True
10    is_sorted([1, 3, 2, 4, 5]) → False
11    is_sorted([1, 2, 3, 4, 5, 6]) → True
12    is_sorted([1, 2, 3, 4, 5, 6, 7]) → True
13    is_sorted([1, 3, 2, 4, 5, 6, 7]) → False
14    is_sorted([1, 2, 2, 3, 3, 4]) → True
15    is_sorted([1, 2, 2, 2, 3, 4]) → False
16    """
```

Instruction 3

```
1 def is_sorted(lst):
2     """
3     Create a function called 'is_sorted' to determine if a
4     list of non-negative integers is in ascending order.
5     The function should also return False if any integer
6     in the list occurs more than twice.
7     """
```

26. In a few words, what is the task this function aims to implement? *

27. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

28. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

29. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 8:

Below are instructions for implementing the function:

Context

```
1 import random
2 class Snake:
3     """
4     The class is a snake game, with allows snake to move and eat food, and also enables to reset,
5     and generat a random food position.
6     """
7
8     def __init__(self, SCREEN_WIDTH, SCREEN_HEIGHT, BLOCK_SIZE, food_position):
9         """
10        Initialize the length of the snake, screen width, screen height, block size,
11        snake head position, score, and food position.
12        :param SCREEN_WIDTH: int
13        :param SCREEN_HEIGHT: int
14        :param BLOCK_SIZE: int, Size of moving units
15        :param food_position: tuple, representing the position(x, y) of food.
16        """
17        self.length = 1
18        self.SCREEN_WIDTH = SCREEN_WIDTH
19        self.SCREEN_HEIGHT = SCREEN_HEIGHT
20        self.BLOCK_SIZE = BLOCK_SIZE
21        self.positions = [((SCREEN_WIDTH / 2), (SCREEN_HEIGHT / 2))]
22        self.score = 0
23        self.food_position = food_position
24
25    def move(self, direction):
26        pass
27
28    def random_food_position(self):
29        pass
30
31    def eat_food(self):
32        pass
33
34
35
36
```

Instruction 1

```
1 def reset(self):
2     """
3     Initialize the snake to its starting condition by setting
4     'self.length' to 1, placing the snake's head at the center
5     using 'SCREEN_WIDTH' and 'SCREEN_HEIGHT', resetting 'self.score'
6     to 0, and calling a method to create a new position for the food
7     that does not overlap with the snake's current position.
8     :return: None
9     """
```

Instruction 2

```
1 def reset(self):
2     """
3     Reset the snake to its initial state. Set the length to 1,
4     the snake head position to ((SCREEN_WIDTH/2), (SCREEN_HEIGHT/2)),
5     the score to 0, and randomly generate new food position.
6     :return: None
7     >>> snake = Snake(100, 100, 1, (51, 51))
8     >>> snake.reset()
9     self.length = 1
10    self.positions = [(50, 50)]
11    self.score = 0
12    self.random_food_position()
13    """
```

Instruction 3

```
1 def reset(self):
2     """
3     Reset the snake to its initial state by setting the
4     "self.length" attribute to 1, updating the "self.positions"
5     list to only include the starting position at the center
6     of the screen using coordinates
7     "((self.SCREEN_WIDTH / 2), (self.SCREEN_HEIGHT / 2))",
8     resetting the "self.score" to 0, and calling the
9     "random_food_position" method to ensure the new food
10    position does not coincide with the snake's body.
11    :return: None
12    """
```

30. In a few words, what is the task this function aims to implement? *

31. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

32. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

33. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 9:

Below are instructions for implementing the function:

Instruction 1

```
1 def circular_shift(x, shift):
2     """
3     Implement a function 'circular_shift' which receives two arguments,
4     an integer 'x' and a shift magnitude 'shift'. The purpose of this
5     function is to execute a rightward circular shift on 'x's digits by
6     'shift' and return this as a string. When 'shift' is more than the
7     number of digits in 'x' (converted to string 's' by 'str(x)'),
8     return a reversed version of 's' ('s[::-1]'). Otherwise, set 'shift'
9     with 'shift %= len(s)'. If this adjusted 'shift' equals 0, then return
10    's' unaltered. Otherwise, stitch together the substring from the last 'shift'
11    positions with the first segment up to
12    'len(s) - shift' to form 's[len(s) - shift:] + s[:len(s) - shift]'.
13
14    """
```

Instruction 2

```
1 def circular_shift(x, shift):
2     """
3     Circular shift the digits of the integer x,
4     shift the digits right by shift
5     and return the result as a string.
6     If shift > number of digits, return digits reversed.
7     >>> circular_shift(12, 1)
8     "21"
9     >>> circular_shift(12, 2)
10    "12"
11    """
```

Instruction 3

```
1 def circular_shift(x, shift):
2     """
3     Write a function named 'circular_shift' which takes two parameters:
4     an integer 'x' and a shift value 'shift'.
5     The function aims to circularly shift the digits of the integer right
6     by the specified number of places and returns the result as a string.
7     If the shift value is greater than the number of digits in the integer,
8     the function returns the digits in reverse order. The function first
9     converts the integer to a string to handle the digits individually.
10    It then applies a modulo operation to the shift value with the length
11    of the string to ensure that the shift doesn't exceed the number of digits.
12    Depending on the shift value, the function rearranges the string accordingly
13    and returns the modified string.
14    """
```

34. In a few words, what is the task this function aims to implement? *

35. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

36. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

37. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 10:

Below are instructions for implementing the function:

Context

```
1 class Warehouse:
2     """
3     The class manages inventory and orders, including adding products,
4     updating product quantities, retrieving product quantities,
5     creating orders, changing order statuses, and tracking orders.
6     """
7
8     def __init__(self):
9         """
10        Initialize two fields.
11        self.inventory is a dict that stores the products.
12        self.inventory = {Product ID: Product}
13        self.orders is a dict that stores the products in a order.
14        self.orders = {Order ID: Order}
15        """
16        self.inventory = {} # Product ID: Product
17        self.orders = {} # Order ID: Order
18
19    def update_product_quantity(self, product_id, quantity):
20        pass
21
22    def get_product_quantity(self, product_id):
23        pass
24
25    def create_order(self, order_id, product_id, quantity):
26        pass
27
28    def change_order_status(self, order_id, status):
29        pass
30
31    def track_order(self, order_id):
32        pass
33
34
35
```

Instruction 1

```
1 def add_product(self, product_id, name, quantity):
2     """
3     Insert a product into 'self.inventory'
4     using values 'product_id', 'name', and 'quantity'.
5     Should the 'product_id' be found in the inventory,
6     augment its 'quantity' with the given amount.
7     If absent, register a new product with these details.
8     :param product_id: int
9     :param name: str, product name
10    :param quantity: int, product quantity
11    """
```

Instruction 2

```
1 def add_product(self, product_id, name, quantity):
2     """
3     Add product to inventory and plus the quantity
4     if it has existed in inventory.
5     Or just add new product to dict otherwise.
6     :param product_id: int
7     :param name: str, product name
8     :param quantity: int, product quantity
9     >>> warehouse.add_product(1, "product1", 3)
10    >>> warehouse.inventory
11    {1: {'name': 'product1', 'quantity': 3}}
12    """
```

Instruction 3

```
1 def add_product(self, product_id, name, quantity):
2     """
3     Incorporate a new item into 'self.inventory'
4     using identifiers 'product_id', 'name', and 'quantity'.
5     If 'product_id' is recognized in the inventory,
6     enhance its quantity by the stipulated additional amount.
7     If unrecognized, initiate a new item in the inventory system.
8     :param product_id: int
9     :param name: str, product name
10    :param quantity: int, product quantity
11    """
```

38. In a few words, what is the task this function aims to implement? *

39. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

40. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

41. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 11:

Below are instructions for implementing the function

Instruction 1

```
1 def is_multiply_prime(a):
2     """
3     Develop a function named 'is_multiply_prime' that checks
4     if a number is composed of the multiplication of exactly
5     three unique prime numbers. The number provided will
6     always be less than 100.
7     """
```

Instruction 2

```
1 def is_multiply_prime(a):
2     """
3     Write a function that returns true if the given number is
4     the multiplication of 3 prime numbers and false otherwise.
5     Knowing that (a) is less than 100.
6     Example:
7     is_multiply_prime(30) == True
8     30 = 2 * 3 * 5
9     """
```

Instruction 3

```
1 def is_multiply_prime(a):
2     """
3     Construct a function called 'is_multiply_prime' that
4     determines if a number is formed by multiplying exactly
5     three different prime numbers. The input value will
6     always be below 100.
7     """
```

42. In a few words, what is the task this function aims to implement? *

43. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

44. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

45. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 12:

Below are instructions for implementing the function:

Context

```
1 class PersonRequest:
2     """
3     This class validates input personal information data and sets
4     invalid fields to None based to specific rules.
5     """
6
7     def __init__(self, name: str, sex: str, phoneNumber: str):
8         """
9         Initialize PersonRequest object with the provided information.
10        :param name: str, the name of the person
11        :param sex: str, the sex of the person
12        :param phoneNumber: str, the phone number of the person
13        """
14        self.name = self._validate_name(name)
15        self.sex = self._validate_sex(sex)
16        self.phoneNumber = self._validate_phoneNumber(phoneNumber)
17
18
19    def _validate_name(self, name: str) -> str:
20        pass
21
22    def _validate_sex(self, sex: str) -> str:
23        pass
24
25
26
```

Instruction 1

```
1 def _validate_phoneNumber(self, phoneNumber: str) -> str:
2     """
3     Examine and return the value of 'phoneNumber'.
4     If the 'phoneNumber' is empty or not precisely 11 digits in length,
5     return 'None'. Initially, the function checks if 'phoneNumber' is
6     missing or is an empty string, and outputs 'None' if it is. Next,
7     it inspects whether the length of 'phoneNumber' is 11, and confirms
8     that every character in 'phoneNumber' is numeric, using
9     'phoneNumber.isdigit()'. If either condition fails, 'None' is returned.
10    If both are true, the provided 'phoneNumber' is returned as valid.
11    :param phoneNumber: str, the phone number to validate
12    :return: str, the validated phone number or None if invalid
13    """
```

Instruction 2

```
1 def _validate_phoneNumber(self, phoneNumber: str) -> str:
2     """
3     Validate the phone number and return it. If phoneNumber is
4     empty or not an 11 digit number, set to None.
5     :param phoneNumber: str, the phone number to validate
6     :return: str, the validated phone number or None if invalid
7     """
```

Instruction 3

```
1 def _validate_phoneNumber(self, phoneNumber: str) -> str:
2     """
3     Validate the phone number "phoneNumber" and return it.
4     If "phoneNumber" is empty or not an 11-digit number, set to "None".
5     :param phoneNumber: str, the phone number to validate
6     :return: str, the validated phone number or None if invalid
7     """
```

46. In a few words, what is the task this function aims to implement? *

47. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

48. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

49. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Demographics Information

50. What is your highest education qualification? *

Une seule réponse possible.

- ☐ Ph.D.
- ☐ Master's Degree
- ☐ Bachelor Degree
- ☐ Autre :

51. What is your field of work? (Engineer, Researcher, Developer, etc.) *

52. How many years of development experience do you have? *

Une seule réponse possible.

- ☐ Less than 5
- ☐ Between 5 and 10
- ☐ More than 10

53. Thank you for completing the survey! Any feedback you would like to provide?

Google Forms

